

VMC-3000 Series | MID 140 (CNC) | MID 142 (ELC) | MID 146 (THM)

|                 |             |                |                                    |
|-----------------|-------------|----------------|------------------------------------|
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**DANGER:** This document contains information on high-voltage circuits (400 V AC, 600 V DC). Electrical work must only be performed by a qualified electrician. Always perform Lockout/Tagout and verify absence of voltage before accessing the cabinet. DC bus capacitors retain hazardous charge for up to 5 minutes after power-off. Never open the drive DC bus covers without first waiting 5 minutes and verifying DC bus voltage <50 V with a calibrated meter.

## 1. Electrical Cabinet Layout

The IP54 steel electrical cabinet (600 mm W × 800 mm H × 400 mm D) is mounted on the rear of the VMC-3000 machine column. The cabinet contains (from top to bottom): incoming isolator, main contactor, servo drive stack, spindle drive, DC bus capacitor bank, 24 V DC PSU, UPS module, PLC I/O rack, CNC controller, terminal strips, and cable entry glands.

| Row     | Component        | Part Number     | Rating          | Description                                    |
|---------|------------------|-----------------|-----------------|------------------------------------------------|
| 1 (Top) | Main Isolator    | ELC-ISO-80A     | 80 A, 3-pole    | Lockable rotary isolator, DIN rail             |
| 1       | Main Contactor   | ELC-CTR-63A     | 63 A, AC3       | Electrically held, auxiliary contacts          |
| 1       | Surge Arrester   | ELC-SPD-B+C     | Type B+C        | 3-phase + N + PE, 25 kA                        |
| 2       | Servo Drive — X  | AXS-DRV-15      | 1.5 kW          | X-axis servo drive                             |
| 2       | Servo Drive — Y  | AXS-DRV-15      | 1.5 kW          | Y-axis servo drive                             |
| 2       | Servo Drive — Z  | AXS-DRV-22      | 2.2 kW          | Z-axis servo drive                             |
| 3       | Spindle Drive    | SPN-DRV-22KW    | 22 kW           | Regenerative spindle inverter                  |
| 3       | DC Bus Capacitor | ELC-CAP-DCB     | 590 V / 4700 uF | Shared DC bus buffer                           |
| 4       | 24 V DC PSU      | ELC-PSU-24V-20A | 24 V / 20 A     | Control voltage supply                         |
| 4       | UPS Module       | ELC-UPS-24V     | 24 V / 10 Ah    | 20 min backup for CNC memory                   |
| 5       | PLC — CPU        | PLC-CPU-500     | —               | EquipmentIQ PLC-500 with CAN master            |
| 5       | PLC — I/O (DI)   | PLC-DI-32       | 24 V DC sink    | 32× digital inputs                             |
| 5       | PLC — I/O (DO)   | PLC-DO-16       | 24 V DC 2A      | 16× digital outputs                            |
| 6       | CNC Controller   | CNC-500         | —               | EquipmentIQ CNC-500, 10 MB program memory      |
| 6       | OPC-UA Gateway   | EIQ-EDGE-001    | —               | Edge gateway, 2 kHz sensor data, OPC-UA server |
| 7 (Bot) | Terminal Blocks  | ELC-TB-4mm2     | —               | Phoenix Contact, 4 mm2 DIN rail                |

Table 1-1: Electrical Cabinet Component Layout

## 2. Power Distribution Wiring

Table 2-1: Power Distribution Wiring

| Signal Name         | From (Module:Pin) | To (Module:Pin) | Wire Gauge | Color Code | Notes                  |
|---------------------|-------------------|-----------------|------------|------------|------------------------|
| L1 Supply (Phase R) | MAINS:L1          | ELC-ISO-80A:1   | 16 mm2     | Brown      | From site distribution |
| L2 Supply (Phase S) | MAINS:L2          | ELC-ISO-80A:3   | 16 mm2     | Black      |                        |
| L3 Supply (Phase T) | MAINS:L3          | ELC-ISO-80A:5   | 16 mm2     | Grey       |                        |

| Signal Name                | From (Module:Pin) | To (Module:Pin) | Wire Gauge | Color Code   | Notes                         |
|----------------------------|-------------------|-----------------|------------|--------------|-------------------------------|
| PE (Protective Earth)      | MAINS:PE          | CAB-PE-BAR      | 16 mm2     | Green/Yellow | Must be <1 ohm to earth       |
| Contactor coil A1          | ELC-PSU-24V:+     | ELC-CTR-63A:A1  | 1.5 mm2    | Red          | Via PLC-DO:Y0.0               |
| DC Bus + to drives         | SPN-DRV-22KW:DC+  | AXS-DRV-X:DC+   | 10 mm2     | Red (DC)     | Shared DC bus link bar        |
| DC Bus - to drives         | SPN-DRV-22KW:DC-  | AXS-DRV-X:DC-   | 10 mm2     | Blue (DC)    |                               |
| 24 VDC Bus +               | ELC-PSU-24V:+24V  | 24V-DIST-BAR:+  | 2.5 mm2    | Red          | Feeds PLC, sensors, solenoids |
| 24 VDC Bus 0V              | ELC-PSU-24V:0V    | 0V-DIST-BAR     | 2.5 mm2    | Blue         |                               |
| UPS bypass                 | ELC-UPS-24V:OUT+  | CNC-500:24V-IN  | 1.5 mm2    | Orange       | CNC always on UPS             |
| Ground bond — machine bed  | CAB-PE-BAR        | MACH-BED:PE     | 16 mm2     | Green/Yellow | <0.1 ohm                      |
| Ground bond — coolant tank | CAB-PE-BAR        | COOL-TANK:PE    | 6 mm2      | Green/Yellow |                               |

### 3. CNC Controller & PLC I/O Mapping

#### 3.1 PLC Digital Inputs (DI)

| Address | Signal Name     | Source                     | Type             | Description                   |
|---------|-----------------|----------------------------|------------------|-------------------------------|
| X0.0    | E-STOP-OP-PANEL | Operator panel E-stop      | NC, safety relay | Main e-stop chain             |
| X0.1    | E-STOP-REAR     | Rear panel E-stop          | NC, safety relay |                               |
| X0.2    | GUARD-DOOR-1    | Main guard door switch     | NC, key switch   | Interlocked to spindle enable |
| X0.3    | GUARD-DOOR-2    | Chip conveyor door         | NC               |                               |
| X0.4    | SPN-ORIENT-SW   | Spindle orientation switch | NPN NO           | P006 orientation confirm      |
| X0.5    | ATC-ARM-HOME    | ATC arm home position      | NPN NO           |                               |
| X0.6    | ATC-TOOL-CLAMP  | Tool clamp confirmed       | NPN NO           | P021 clamp pressure OK        |
| X0.7    | COOLANT-FLOW-SW | Coolant flow switch        | NPN NO           | P030 flow confirm             |
| X1.0    | HOME-X          | X-axis home switch         | NPN NO           |                               |
| X1.1    | HOME-Y          | Y-axis home switch         | NPN NO           |                               |
| X1.2    | HOME-Z          | Z-axis home switch         | NPN NO           |                               |
| X1.3    | OT-X+           | X+ over-travel             | NC hardwired     | In e-stop chain               |
| X1.4    | OT-X-           | X- over-travel             | NC hardwired     |                               |
| X1.5    | OT-Y+           | Y+ over-travel             | NC hardwired     |                               |
| X1.6    | LUB-FLOW-OK     | Lubrication flow switch    | NPN NO           | P040 delivery confirm         |
| X1.7    | HYD-PRESSURE-OK | Hydraulic pressure switch  | NPN NO           | P050 >60 bar confirm          |

Table 3-1: PLC Digital Input (DI) Mapping

#### 3.2 PLC Digital Outputs (DO)

| Address | Signal Name    | Load                      | Description   |
|---------|----------------|---------------------------|---------------|
| Y0.0    | MAIN-CONTACTOR | ELC-CTR-63A coil, 24V 10W | Main power on |

| Address | Signal Name   | Load                            | Description                       |
|---------|---------------|---------------------------------|-----------------------------------|
| Y0.1    | COOLANT-PUMP  | CLS-PUMP relay, 2.2 kW          | Coolant pump run                  |
| Y0.2    | LUB-PUMP      | LUB-PUMP relay, 0.18 kW         | Lubrication pump run              |
| Y0.3    | SPN-BRAKE     | Spindle brake solenoid, 24V 15W | Release spindle orientation brake |
| Y0.4    | HYD-PUMP      | HYD pump relay, 1.5 kW          | Hydraulic pump run                |
| Y0.5    | Z-BRAKE       | Z-axis brake, 24V 30W           | Release Z-axis gravity brake      |
| Y0.6    | ATC-ARM-FWD   | ATC solenoid A, 24V 10W         | ATC arm swing forward             |
| Y0.7    | ATC-ARM-RET   | ATC solenoid B, 24V 10W         | ATC arm swing return              |
| Y1.0    | TOOL-UNCLAMP  | Tool unclamp solenoid, 24V      | Release draw-bar for tool change  |
| Y1.1    | COOLANT-TSC   | TSC solenoid valve, 24V         | Through-spindle coolant valve     |
| Y1.2    | ALARM-LIGHT-R | Stack light red, 24V 5W         | Active alarm indication           |
| Y1.3    | ALARM-LIGHT-Y | Stack light yellow, 24V 5W      | Warning indication                |
| Y1.4    | ALARM-LIGHT-G | Stack light green, 24V 5W       | Machine ready                     |

Table 3-2: PLC Digital Output (DO) Mapping

4. Electrical & CNC Parameters

| Param ID | PID | Parameter Name          | Unit  | Normal Range | Critical Range |
|----------|-----|-------------------------|-------|--------------|----------------|
| P070     | 70  | Ambient Temperature     | °C    | 15 – 35      | 5 – 45         |
| P071     | 71  | Electrical Cabinet Temp | °C    | 20 – 45      | 10 – 60        |
| P072     | 72  | Feed Drive Motor Temp   | °C    | 20 – 72      | 0 – 90         |
| P080     | 80  | CNC CPU Load            | %     | 0 – 75       | 0 – 100        |
| P081     | 81  | CNC Memory Usage        | %     | 0 – 80       | 0 – 100        |
| P082     | 82  | NC Program Block Number | block | 0 – 99999    | 0 – 99999      |
| P083     | 83  | Feed Override           | %     | 0 – 120      | 0 – 150        |
| P090     | 90  | DC Bus Voltage          | V     | 560 – 620    | 480 – 680      |
| P091     | 91  | Main Supply Voltage     | V     | 380 – 420    | 340 – 440      |
| P092     | 92  | Ground Leakage Current  | mA    | 0 – 10       | 0 – 30         |
| P093     | 93  | UPS Battery Level       | %     | 80 – 100     | 20 – 100       |

Table 4-1: ELC/THM/CNC Monitored Parameters

5. Error Codes — ELC, THM, CNC Subsystems

| Error Code | Severity | Title                                     | Primary Param | Action                                      |
|------------|----------|-------------------------------------------|---------------|---------------------------------------------|
| ELC-CR-001 | CRITICAL | DC Bus Undervoltage — All Drives Disabled | —             | Immediate machine stop. Do not restart with |
| ELC-MJ-001 | MAJOR    | Supply Voltage — Out of Tolerance         | —             | Stop current operation. Machine must be in  |

| Error Code | Severity | Title                                         | Primary Param | Action                                                                  |
|------------|----------|-----------------------------------------------|---------------|-------------------------------------------------------------------------|
| ELC-SR-001 | SERIOUS  | Ground Leakage — Elevated                     | —             | Complete current cycle then stop. Schedule maintenance within 24 hours. |
| ELC-MD-001 | MODERATE | UPS Battery — Low Charge                      | —             | Log fault. Schedule maintenance within 24 hours.                        |
| ELC-MN-001 | MINOR    | Electrical Cabinet Temperature — Elevated     | —             | Log fault. Monitor parameters. Schedule maintenance within 24 hours.    |
| ELC-WN-001 | WARNING  | DC Bus — Minor Voltage Ripple Detected        | —             | Log event. No immediate action required. Monitor parameters.            |
| ELC-NC-001 | NOTICE   | Power-On Self-Test — Passed                   | —             | Informational. Parameter approaching limit.                             |
| ELC-AD-001 | ADVISORY | UPS Battery — Fully Charged                   | —             | Performance advisory only. No maintenance required.                     |
| THM-CR-001 | CRITICAL | Multi-Zone Overtemperature — Emergency Stop   | —             | Immediate machine stop. Do not restart without investigation.           |
| THM-MD-001 | MODERATE | Thermal Gradient — Elevated Ambient           | —             | Log fault. Schedule maintenance within 24 hours.                        |
| THM-WN-001 | WARNING  | Ambient Temperature — Above Recommended Range | —             | Log event. No immediate action required. Monitor parameters.            |
| THM-NC-001 | NOTICE   | Thermal Compensation Active                   | —             | Informational. Parameter approaching limit.                             |
| THM-AD-001 | ADVISORY | Feed Motor Temperature — Nominal              | —             | Performance advisory only. No maintenance required.                     |
| CNC-CR-001 | CRITICAL | CNC Controller — Watchdog Timeout             | —             | Immediate machine stop. Do not restart without investigation.           |
| CNC-MJ-001 | MAJOR    | NC Program — Syntax Error at Block            | —             | Stop current operation. Machine must be in manual mode.                 |
| CNC-SR-001 | SERIOUS  | CNC CPU Load — High                           | —             | Complete current cycle then stop. Schedule maintenance within 24 hours. |
| CNC-MD-011 | MODERATE | CNC Memory — Program Storage Near Full        | —             | Log fault. Schedule maintenance within 24 hours.                        |
| CNC-MN-001 | MINOR    | Feed Override — Non-100% Setting              | —             | Log fault. Monitor parameters. Schedule maintenance within 24 hours.    |
| CNC-WN-001 | WARNING  | Program — Approaching End of File             | —             | Log event. No immediate action required. Monitor parameters.            |
| CNC-NC-001 | NOTICE   | CNC System — Normal Operation                 | —             | Informational. Parameter approaching limit.                             |
| CNC-AD-001 | ADVISORY | NC Block Counter                              | —             | Performance advisory only. No maintenance required.                     |

Table 5-1: Electrical, Thermal & CNC Controller Error Codes

## 6. Spare Parts — Electrical Cabinet

| Part Number     | Description                                  | Qty | Unit | Lead Time | Safety Stock |
|-----------------|----------------------------------------------|-----|------|-----------|--------------|
| ELC-ISO-80A     | Main rotary isolator 80A 3-pole lockable     | 1   | Each | 1 week    | 1            |
| ELC-CTR-63A     | Main contactor 63A AC3 with 24V coil         | 1   | Each | 1 week    | 1            |
| ELC-PSU-24V-20A | 24 VDC power supply 20 A DIN rail            | 1   | Each | 1 week    | 1            |
| ELC-UPS-24V     | UPS module 24V 10 Ah (20 min backup)         | 1   | Each | 2 weeks   | 1            |
| ELC-CAP-DCB     | DC bus capacitor 590V 4700uF (matched pair)  | 1   | Pair | 4 weeks   | 1            |
| PLC-CPU-500     | PLC CPU module EIQ-500 with CANopen master   | 1   | Each | 4 weeks   | 1            |
| PLC-DI-32       | PLC digital input module 32ch 24VDC          | 1   | Each | 2 weeks   | 1            |
| PLC-DO-16       | PLC digital output module 16ch 24VDC 2A      | 1   | Each | 2 weeks   | 1            |
| CNC-500         | CNC controller complete (backup replacement) | 1   | Each | 6 weeks   | 1            |
| EIQ-EDGE-001    | EquipmentIQ OPC-UA edge gateway              | 1   | Each | 3 weeks   | 1            |

| Part Number    | Description                                | Qty | Unit | Lead Time | Safety Stock |
|----------------|--------------------------------------------|-----|------|-----------|--------------|
| ELC-FILTER-CAB | Cabinet air filter mat 300x300mm (10 pack) | 1   | Pack | 1 week    | 2            |

Table 6-1: Electrical Cabinet Spare Parts List